



Force modeling of a reluctance motion system with multi-axial asymmetrical air gaps[☆]

Michael Pumphrey^a, Mohammad Al Saaideh^{b,*}, Yazan M. Al-Rawashdeh^b, Natheer Alatawneh^b,
Mohammad Al Janaideh^{a,b,c}

^a School of Engineering, University of Guelph, Guelph, ON N1G 2W1, Canada

^b Department of Mechanical and Mechatronics Engineering, Memorial University, St. John's, NL A1B 3X5, Canada

^c Department of Mathematics, Faculty of Civil Engineering, Czech Technical University, Prague 6, Czech Republic

ARTICLE INFO

Keywords:

Reluctance actuator
Magnetic force
Asymmetrical air gaps
Magnetic flux
Photolithography
Precision motion system

ABSTRACT

Semiconductor photolithography equipment requires an actuation system that can satisfy high acceleration and precision demands on the nanoscale in order to operate with high throughput and efficiency. Reluctance actuators, which offer greater acceleration and force output than conventional Lorentz actuators, are one option. The photolithography technique uses a floating stage with air bearings to reduce friction; nonetheless, vibration transfer is not completely prevented, potentially causing misalignment and asymmetries between the actuator components. The output force can be significantly impacted by unequal offsets between the mover and stator. In the paper that follows, a technique for calculating the force of various asymmetrical instances for the C-core reluctance actuator is demonstrated. This paper looks into the effect of rotational offsets in all rotational degrees of freedom occurring simultaneously. Previously, only each degree of freedom was analyzed. Analytical models are developed and compared to Comsol Multiphysics (COMSOL) simulations, and experimental findings show accuracy to be within almost 10% for the ranges examined. The findings presented can allow for future control systems to be designed to counteract multi-axial asymmetric issues.

1. Introduction

In order to enhance IC throughput, memory storage, and processing speeds in the future, integrated circuit (IC) production will need more sophisticated and optimal control [1,2]. Nanoscale geometric designs are transferred and encoded into the IC using photolithography devices. A floating stage with friction-free air bearings is employed to lessen vibration transfer [1]. The motion platform can hover with the use of these air bearings, preventing any contact that may cause friction on nanoscale motions [3]. The air-bearing generates air vertices, which introduce minor vibrations into the positioning stage [4] and cause self-excited instabilities that could harm the entire positioning stage; therefore vibration transfer is not completely removed [5]. This would result in general positioning errors, which would lower the positioning stage's accuracy [6]. The misalignment would impart asymmetries, which would result in notable changes in force output profiles depending on the asymmetry situation. In most cases, multiple rotational degrees of freedom would be affected to some degree.

High precision allows more geometric features to be encoded onto IC components, which increases the amount of encoded data [7]. It takes a dual-stage actuation mechanism to ensure excellent precision throughout longer strokes [2,8]. This device uses two actuation stages, a short-stroke stage, and a long-stroke stage. The long-stroke stage can generate a vast range of motions up to 1 meter, and the short-stroke stage can then fine-tune the motion on the nanoscale, as shown in Fig. 1. Currently, in both the reticle and wafer phases, Lorentz actuators are mostly used for short-stroke. However, compared to the typical Lorentz actuator, the reluctance actuator can produce higher force density and acceleration [7,9]. To improve motion control further, it is necessary to study the reluctance actuator for every altered case including multi-axial offsets.

Since the input coil current and the output force relationship are highly linearly coupled for Lorentz actuators, they are easier to work with and evaluate [11]. This allows for clear analytical models of the force estimations, allowing the photolithography machines to function across wide operating ranges with great positioning accuracy [12]. The

[☆] This paper was recommended by Associate editor Prof. R. Leach.

* Corresponding author.

E-mail addresses: mpumphre@uoguelph.ca (M. Pumphrey), mialsaaideh@mun.ca (M.A. Saaideh), yalrawashdeh@mun.ca (Y.M. Al-Rawashdeh), nalatawneh@mun.ca (N. Alatawneh), maljanai@uoguelph.ca (M.A. Janaideh).

<https://doi.org/10.1016/j.precisioneng.2023.12.009>

Received 18 November 2023; Received in revised form 18 December 2023; Accepted 19 December 2023

Available online 23 December 2023

0141-6359/© 2023 Elsevier Inc. All rights reserved.

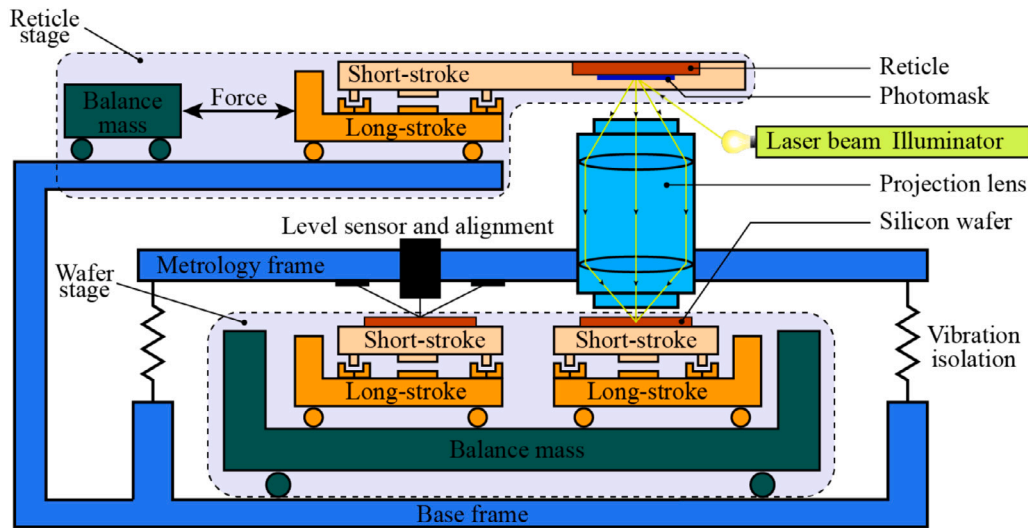


Fig. 1. A schematic of the lithography machine architecture, which is composed of the reticle stage, projection lenses, and wafer stage. Dual-stage long-stroke and short-stroke Lorentz actuators are used on the wafer and reticle stages to correct positioning errors [10].

disadvantage associated with Lorentz actuators is their low force density, which necessitates the use of high currents, resulting in significant energy dissipation in the form of heat [11]. However, the reluctance actuator is much more efficient, and can produce up to 10 times the force density of the Lorentz actuator [11].

Due to the nonlinearity of the current–force and position–force relationships, the reluctance actuator is more difficult to model than the Lorentz actuator. A number of contributions have been made to understanding these nonlinearities in the literature. For example, the nonlinear interactions in [13] are modeled for real-time control applications using feedback from a capacitive displacement sensor, achieving 2 nm resolution and a range of 60 μm .

Additionally, there are numerous documented examples of photolithographic control designs. A synchronization control algorithm, for instance, is created in [14] to enhance synchronization between the reticle and wafer stages. For very precise motion control systems, coupled system identification and robust control are demonstrated in [15]. A data-based multi-input multi-output feedforward control method is applied to the motion control of the wafer stage in [16], which has the capacity to mitigate unidentified disturbances in the motion stage. In [17], feedforward control is used to enhance settling time and reduce crosstalk in the system by utilizing data that is previously known for the wafer stage.

Reluctance actuator modeling and control strategies have been considered in a number of research papers. For instance, in [18], the accuracy of finite element analysis (FEA) is examined by calculating the frequency response of a tip-tilt two degrees of freedom (2-DoF) in both x and y rotations hybrid reluctance actuator under monoaxial excitation and with results being verified through experimental testing. In [19], a flux-controlled hybrid reluctance actuator is investigated. Magnetic flux serves as an input, and sensor fusion from a search coil and current monitor is used to anticipate flux. The nonlinearities are then managed by a Proportional Integral (PI) flux feedback controller. Also, [20] looks into a flux feedback-controlled reluctance actuator with the help of a cascaded analogue sensing coil and a hall probe flux feedback, the reluctance actuator may be linearized and is confirmed experimentally. In addition to the benefit of improved energy efficiency, this offers the reluctance actuator the linear behavior of the Lorentz actuator.

Flux leakage can also be a concern, so it is crucial to have a reliable detection mechanism. A compensation mechanism for magnetic flux leakage measurement errors is created and tested using modeling and actual physical testing in [21]. An experimentally validated run-to-run control for soft landing is provided in [22] and is based on

Bayesian optimization of the input current of a reluctance actuator. The model can potentially be improved by simulating the hysteresis effects of the reluctance actuator. [23] proposes a multilayer adaptive neural network-based nonlinear current control setup with hysteresis compensation. A regressive radial basis function (RBF) neural network model for rate-dependent hysteresis modeling is developed in [24] and performs well in hysteresis estimations for reluctance actuators when compared to other approaches. A current-driven reluctance actuator is linearized using a hysteretic inverse actuator model, which also models the influence of hysteresis in [25]. In [26], two semi-analytic approaches are compared for their ability to forecast magnetic hysteresis in an E-core reluctance actuator output force. An open-loop steady-state model that predicts the hysteresis effects of an E-core actuator based on the phase delay of flux fluctuations is studied in [26]. In [27,28], a hybrid lumped-parameter model that considers the mechanical and electromagnetic dynamics, as well as eddy currents, flux fringing, magnetic hysteresis, and saturation, is presented. In [28] the induced eddy currents and flux fringing effects are modeled using computed Comsol Multiphysics (COMSOL) data. In [29] a resonant rotational reluctance actuator for scanning mirrors is investigated and shown to outperform state-of-the-art systems. In [30], a multi-objective optimization algorithm is proposed for the optimal design of an E-core reluctance actuator to meet the requirements of the precision motion system. The optimization goals include maximizing the output magnetic force, minimizing both the mass and time constant of the actuator, and driving the actuator stiffness value towards zero. The simulation of the dynamic response of the reluctance actuator confirms the effectiveness of the optimal design. In [31], an output feedback control approach based on an extended high-gain observer is proposed for a motion system driven by the reluctance actuator. The proposed observer uses the measured position to estimate the magnetic flux and unknown nonlinearities of the reluctance actuator, providing the state feedback controller with this information to achieve tracking to a desired motion profile.

2. Contributions

This paper extends the precursor [10] by having asymmetrical offsets in all angular degrees of freedom, as shown in Fig. 2. In the previous paper, analytical models for the force for each rotational degree of freedom were found individually with assumptions of linear flux trajectories which were further refined with a polynomial correction function to better match the simulated results and did not analyze

when the reluctance actuator features multiple rotations in each axis simultaneously. The main objective is to devise a complete output force analytical modeling for a reluctance actuator that is offset rotationally in multiple axes simultaneously. This paper provides a method to obtain such forces, evaluating the influence that each asymmetrical case has and combining their influences together to generate a final mathematical equation for force estimation. The main knowledge gap here is the difficulty of dealing with the control of motion of the reluctance actuator in high precision due to asymmetry in the air gap between the mover and the stator. Specifically, under multi-axial asymmetrical air gaps situations in motion control. This study derived an analytical equation to describe such output forces within multi-axial asymmetrical air gaps. We derived and studied these output forces considering FEM model to improve the model accuracy. We also tested the model experimentally to verify the model under multi-axial asymmetrical air gaps. The proposed method will be beneficial for rotational-based reluctance actuated systems for defining the output force at specific multi-axial offsets. Currently, rotational reluctance actuators can be used in resonant laser scanner applications [29] and can provide state-of-the-art performance. With better force modeling of the effect of multiple angular offsets, better torque output equations can be found for rotational reluctance actuators that offer multi-axial outputs to expand the state of the art further. The method may also help other fields where force equations need to be combined by highlighting their effective influence on one another. The contribution of the paper can be summarized as

- Derive an analytical equation that describes the output force of the reluctance actuator under multi-axial asymmetrical air gaps. We derived this equation and improved its accuracy by fitting the equation using FEM data.
- Implement a FEM setup to further tune the analytical output force equation via a polynomial curve fitted correction function.
- Experiment on multi-axial offset cases for the reluctance actuator with the use of a 3D printed modular enclosure to house various offset cases to verify and confirm the analytical model.

The applications that the force characterization of multi-axial reluctance actuators can be applied to are:

- Force characterization of undesired axial offset disturbances from reluctance actuators operated under air-bearing in semiconductor lithography manufacturing machines. Force characterization will benefit overall error budgeting calculations and could help in control design of such actuators to understand disturbance forecasting and increase performance.
- Rotational reluctance actuators where the mover is meant to be axially offset in various degrees of freedom depending on the application such as laser optics and precision scanning. The use of asymmetries can be taken advantage of to allow for a force and torque to be an output of the system which can achieve the desired position and angular specifications depending on the requirements of the user allowing more advanced micropositioning actuator outputs to be realized.

The paper is outlined as follows: a combination of angular offset cases to form a general multi-axial asymmetrical case in Section 3. The COMSOL FEA models are built and compared to the analytical model in Section 4. In Section 5, the experiment is documented with comparisons to the analytical results, with a discussion presented in Section 6. Lastly, Section 7 presents the conclusions of this study, and future work is presented.

3. Analytical model

In the predecessor [10] to this paper, the angular disturbances were found on an individual angular degree of freedom basis, as shown in

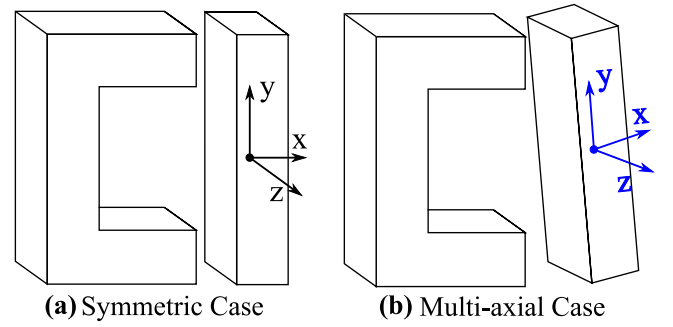


Fig. 2. (a) The standard case with symmetric air gaps. (b) The multi-axial case with asymmetric air gaps.

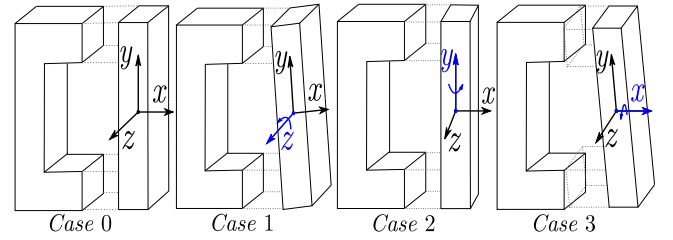


Fig. 3. Case 0 is the ideal case with symmetrical air gaps. Case 1, 2, and 3, have angular disturbances in the z, y, and x axis respectively [10].

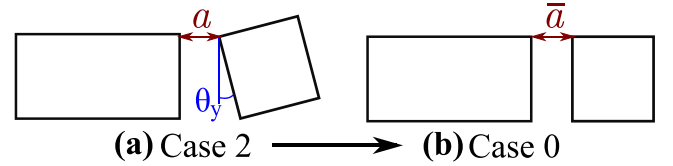


Fig. 4. The effective air gap needs to be found that will generalize (a) Case 2 into the standard (b) Case 0.

Fig. 3. Another paper that investigated an individual angular degree of freedom for a reluctance actuator is [29] which uses a similar approach as [10], along with using a polynomial approximation based on FEA results. By integrating the gap length along the edge, it is possible to identify the reluctance for a non-parallel gap [10]. The force can only be roughly predicted by the analytical model, which is sometimes inaccurate due to non-accounted saturation effects. FEA modeling is used to improve the force estimates through curve fitting of the FEA results to produce a correction function as seen in [10,29].

For Case 1 there would be two separate forces on the top and bottom of the I-beam, whereas the other cases would have the same forces due to the geometry in each air gap. The first step to generalizing a combination of all the cases together was to find the influence that each case has on the original Case 0. To begin, Case 2 was examined first to find the effect the angle has on the effective air gap, meaning what air gap would produce the same force if used in Case 0 as shown in Fig. 4.

Case 2 and Case 0 can be compared analytically to determine the effect the angle has on the effective air gap, $\bar{a}$. The force, F_1 , and the reluctance, R_1 , in each air gap is found to be [10]:

$$F_1 = \frac{N^2 i^2 R_1^2 \mu_0 A^*}{2 R_T^2 a (b \tan \theta_y + a)} \quad (1)$$

$$R_1 = \frac{b \tan \theta_y}{\mu_0 A \ln \left(\frac{b \tan \theta_y}{a} + 1 \right)} \quad (2)$$

where μ_0 is the permeability of free space, R_T is the total reluctance in the magnetic circuit, N is the number of coil turns, i is the coil current, and a is the air gap displacement. θ_y is the angular displacement along the y axis of the I-beam, A is the original area and A^* is A multiplied

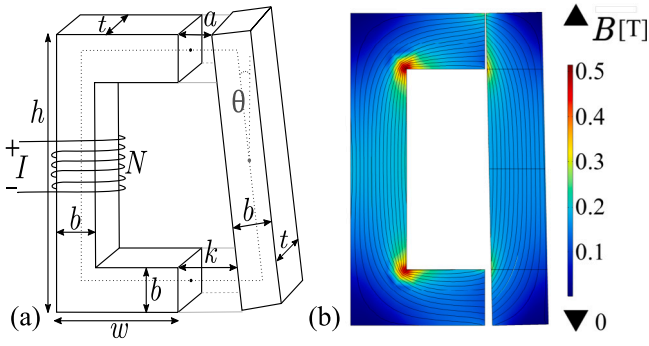


Fig. 5. Case 1, (a) Schematic representation of the C-Core reluctance actuator and (b) COMSOL flux density distribution FEA of C-Core [10].

by the polynomial correction factor found in [10], b is the width as described in Fig. 5.

The reluctance in the C-Core, R_C , and the I-Beam, R_I , is found

$$R_C = \frac{h + 2(w - b)}{\mu_r \mu_0 A}, \quad R_I = \frac{h}{\mu_r \mu_0 A}. \quad (3)$$

The total reluctance of the system, R_T is found

$$R_T = R_1 + R_2 + R_C + R_I \quad (4)$$

For Cases 0, 2, and 3 the reluctance in both top R_1 and bottom R_2 are both equal $R_1 = R_2$. Case 1 is the only case where $R_1 \neq R_2$ due to the differing air gap displacements in both top and bottom for this case. This means the force is different in the top and bottom air gaps for Case 1 setups.

Whereas in Case 0 the force, F_1 , and the reluctance, R_1 , in the air gap is:

$$F_1 = \frac{N^2 i^2 R_1^2 \mu_0 A}{2 R_T^2 a^2} \quad (5)$$

$$R_1 = \frac{a}{\mu_0 A} \quad (6)$$

Through comparison, Case 2 can be represented as Case 0 by augmenting the effective reluctance and air gap contribution. Comparing Eq. (1) with Eq. (5) the effective air gap is: $\bar{a} = \sqrt{a(b \tan \theta_y + a)}$ and the area is augmented to A^* with the derived polynomial correction factor from [10]. The reluctance equation is augmented by switching the air gap, a , by $a_R = (b \tan \theta_y) / (\ln(b \tan \theta_y / a + 1))$.

3.1. Case 1 + 2

Since Case 2 has been modeled to Case 0 and since Case 1 is based on Case 0, the effective air gap, $\bar{a}$, and the influence on the reluctance can be introduced into Case 1 to form Case 1 + 2. Cases 1 and 2 can be combined since Case 2 affects the effective air gap of both top and bottom equally based on the angular offset. This effective air gap can then be introduced in the reluctance and force equations for Case 1 which creates the combination of Case 1 + 2 as shown in Fig. 6.

The new equation would combine the polynomial correction functions on the effective area of Case 1 and Case 2 to get the new effective area, A_1^* , for the top air gap and, A_2^* , for the bottom air gap. The effective air gap, $\bar{a}$, is used and the effect of the air gap on the reluctance equation, a_R , is used in the reluctance equation. This forms the general equation for Case 1 + 2:

$$F_1 = \frac{N^2 i^2 R_1^2 \mu_0 A_1^*}{2 R_T^2 \bar{a} (b \tan \theta_z + \bar{a})} \quad (7)$$

$$F_2 = \frac{N^2 i^2 R_2^2 \mu_0 A_2^*}{2 R_T^2 \bar{k} (b \tan \theta_z + \bar{k})} \quad (8)$$

$$R_1 = \frac{b \tan \theta_z}{\mu_0 A \ln \left(\frac{b \tan \theta_z}{a_R} + 1 \right)} \quad (9)$$

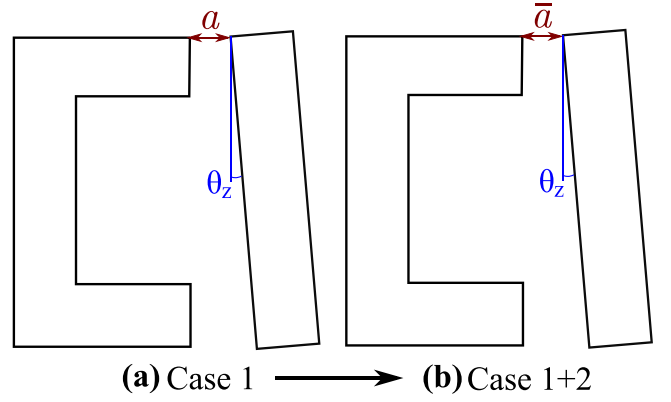


Fig. 6. (a) Case 1 can be converted to the combination of Case 1 + 2 with the introduction of the effective air gap, $\bar{a}$, along with the derived polynomial correction factors and influence on the reluctance.

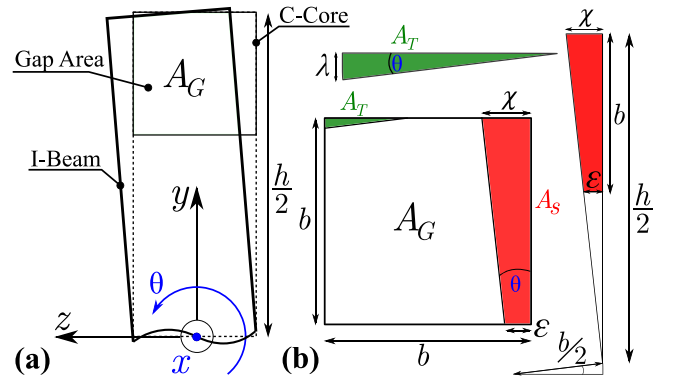


Fig. 7. Geometric representation (a) of the tilted cross-section for the Case 3 reluctance derivation, with a rotation in the x -axis of the I-beam. The air gap cross-section (b) is shown to not be square due to the tilted I-beam [10].

$$R_2 = \frac{b \tan \theta_z}{\mu_0 A \ln \left(\frac{b \tan \theta_z}{k_R} + 1 \right)}, \quad (10)$$

where $\bar{k} = \bar{a} + (h - b) \tan \theta_z$ and $k_R = a_R + (h - b) \tan \theta_z$. θ_z is the angular offset in the z -axis.

3.2. Case 1 + 2 + 3

Finally, all three augmented cases can be combined. The effect Case 3 has is to augment the overall air gap by A_G as outlined in Fig. 7 [10]. When there is a small air gap the magnetic flux takes the path of least reluctance which is the overlapped area for Case 3 as the mover is close to the stator in this region. For a large gap, flux can flow through the un-overlapped region as there is not much difference in the reluctance between the overlapped and un-overlapped region due to the large gap/path that the flux has to take regardless. The polynomial correction fixes this issue.

This would lead to the final augmented area, A_1^* , for the top air gap and, A_2^* for the bottom air gap, which is the product of the polynomial correction functions for each rotational degree of freedom multiplied by A_G . The equations for A_1^* and A_2^* are found in Appendix.

Leading to the final force equations for Case 1 + 2 + 3:

$$F_1 = \frac{N^2 i^2 R_1^2 \mu_0 A_1^*}{2 R_T^2 \bar{a} (b \tan \theta_z + \bar{a})} \quad (11)$$

$$F_2 = \frac{N^2 i^2 R_2^2 \mu_0 A_2^*}{2 R_T^2 \bar{k} (b \tan \theta_z + \bar{k})} \quad (12)$$

$$F_T = F_1 + F_2 \quad (13)$$

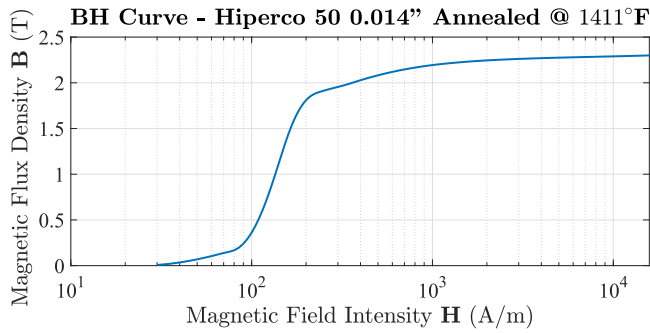


Fig. 8. BH curve for the Hipercro 50 0.014" annealed at 1411 °F in which the I-beam and C-core are comprised of. This curve is used in the simulation and analytical models [10].

Table 1

Setup trials to simulate and compare with the analytic solution for verification.

Setup #	θ_x (mrad)	θ_y (mrad)	θ_z (mrad)
1	5	5	5
2	10	10	2
3	25	20	5
4	0	25	2
5	15	0	5

where F_T is the combined force of the top and bottom air gaps. Now this solution can be compared with COMSOL results in the following section for verification.

4. Simulation results

The reluctance actuator was constructed and tested using COMSOL, a finite element analysis (FEA) multiphysics modeling software engine, in order to validate the analytical model. In [27,28], a similar strategy was used, utilizing FEA to identify physics correlations for complicated systems in order to derive approximate governing equations. Based on the preceding schematic in Fig. 5a, the reluctance actuator under investigation has dimensions of $b = t = 25$ mm, $w = 60$ mm, $h = 140$ mm, and $N = 400$ turns for all simulation and experimental models.

The BH-curve of the core material, also known as the magnetic field density versus magnetic field intensity curve, was used, as illustrated in Fig. 8. Using a lookup table of the material's provided BH-curve, the analytical model can subsequently be adjusted to consider this saturation. The model material consists of Hipercro 50, annealed at 1411 °F, which is an iron-cobalt vanadium alloy and the BH-curve can be seen in Fig. 8.

The model was tested with multiple misalignment cases between the stator and mover, as seen in Table 1. An example of the COMSOL simulation is shown in Fig. 9 of Setup #3.

When compared to the analytic solution for air gaps ranging from 0.5 mm to 1.5 mm it was found to accurately match the force within ~5% for most scenarios as seen in Fig. 10. For cases with $\theta_z = 2$ mrad, the error was under 3%, showing the solution works best with lower θ_z angles. It was found that θ_z had the most influence on the error, which makes sense since it has the most overall effect on the total force when comparing the standard cases as seen in Fig. 11 [10]. Next, the analytical solution is compared to an experimental setup of the reluctance actuator as described in the following section.

5. Experimental design and procedure

Two enclosures for the I-beam and C-Core as well as a force bracket were designed in SolidWorks and 3D printed in order to study the forces acting on the beam at various air gaps. The I-beam and the C-Core will be housed and isolated within these enclosures, enabling static

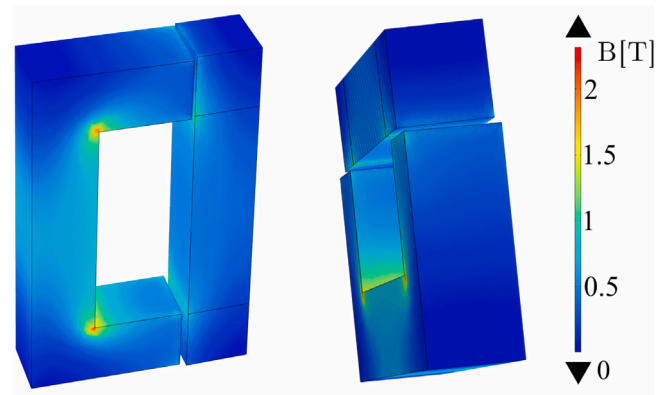


Fig. 9. COMSOL model for Setup #3 which features $\theta_x = 25$ mrad, $\theta_y = 20$ mrad, and $\theta_z = 5$ mrad.

measurements of the force created at various spacing. Fig. 12 depicts these enclosures. The experimental setup can be seen in Fig. 13. The experiment is conducted by applying an electrical current to the coil at specific air gap displacements and recording the force output of the four sensors.

1A DC power source with a maximum voltage and current of 30 V and 5 A, respectively, is used to operate the reluctance actuator. The stator and mover are magnetically attracted, causing the I-core enclosure to be pushed against the fixed base. The four compression sensors (FX293X-100A-0010-L) located in the corners of the F-Bracket are pressed against the I-box dimple extrusions as it is forced against the F-Bracket. A dSPACE data board (DS1104) reads the output voltage from the force sensors. The magnetic force can be calculated from the output voltage. The sensitivity and offset needed based on the voltage readings were determined by calibrating each of the four sensors separately with different weights and plotting the results. Each sensor's sensitivity was somewhat varied, measuring 114.05, 115.38, 114.48, 114.91 mV/N, with an offset of -5.53, -5.25, -5.09, and -5.51 mV respectively.

The procedure is to first embed the I-core into the modular I-box enclosure which fixes the angles to that of a specific case. The I-box now housing the I-core is seated over the fixed C-box which houses the C-core. The two coils each consist of $N = 200$ windings of copper wire. The coils are seated on each tooth of the C-core. The C-box is fixed to a vibration resistant granite table via bolts. The air gap displacement between the I-beam and the C-core is measured using electronic calipers. The experiment is then conducted by altering the electric current and measuring the force response for five different multi-axial cases.

Layered lamination of annealed "Hipercro 50" iron-cobalt vanadium alloy was used to construct the I-beam and C-Core. A modular I-box design with interchangeable backing that can be 3D printed was created to position the I-beam in various multi-axial angular disturbance instances, as shown in Fig. 14. The experimental findings compared to the analytical model are shown in Fig. 15. This shows that the analytical model can be used to evaluate forces when multi-axial offsets are considered and are within ~10%.

6. Discussion

For each case, it was discovered that the experimental model matched the analytical equations that FEA and analysis had produced with an error rate under ~10%. This experimental setup was fabricated completely in the lab so there would be errors due to 3D printing tolerances, laminations, sensor integration and location, and coil windings. The experiment was also performed under normal laboratory conditions (some environmental disturbances such as noise and temperature can imbue random errors in the setup). All these potential errors

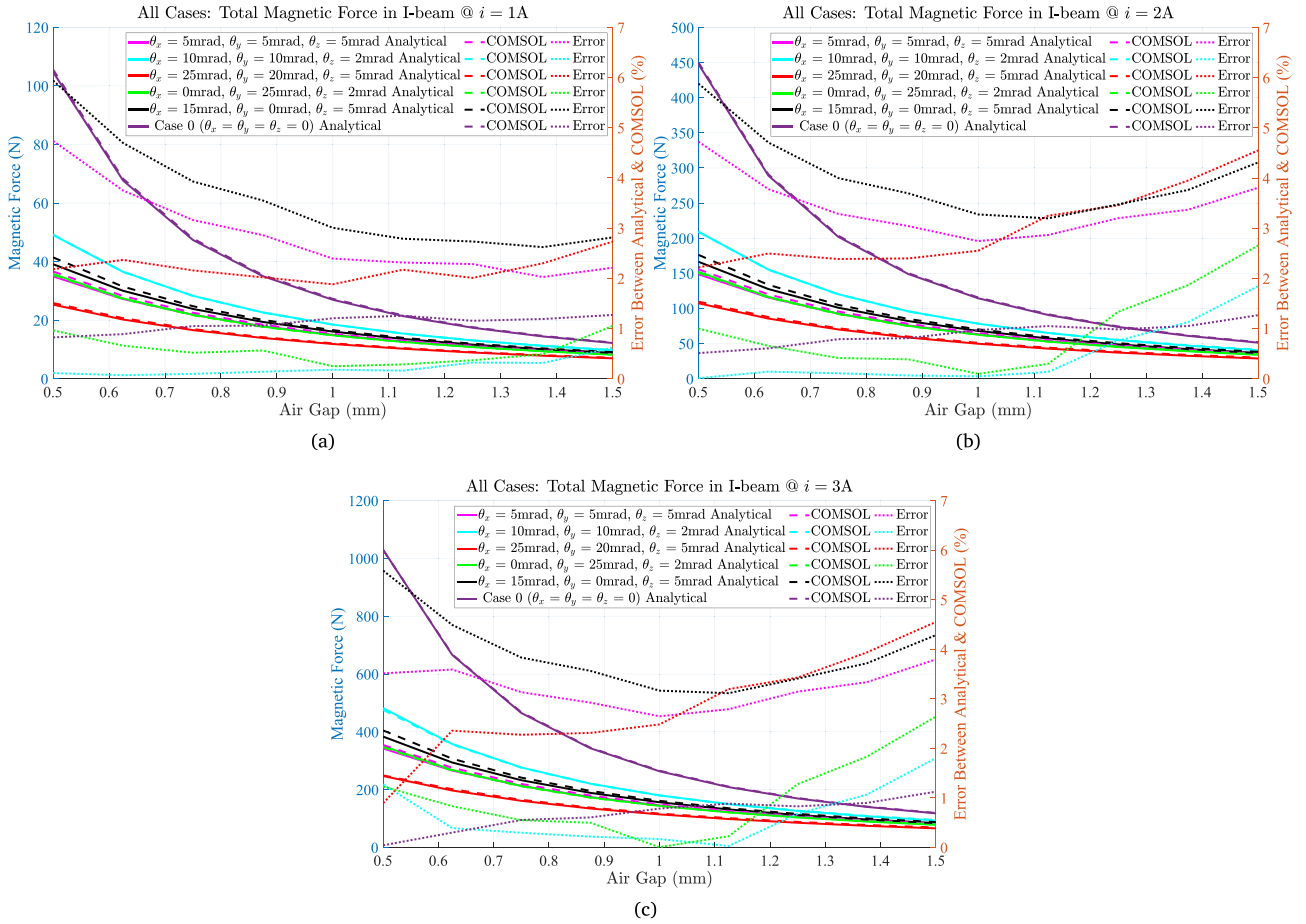


Fig. 10. The analytical and COMSOL models for the multi-axis Case 1 + 2 + 3, are shown for (a) $i = 1$ A, (b) $i = 2$ A, and (c) $i = 3$ A to match with a maximum error of (a) 5.9%, (b) 5.8%, and (c) 5.6% for a range of 0.5–1.5 mm of air gap displacement.

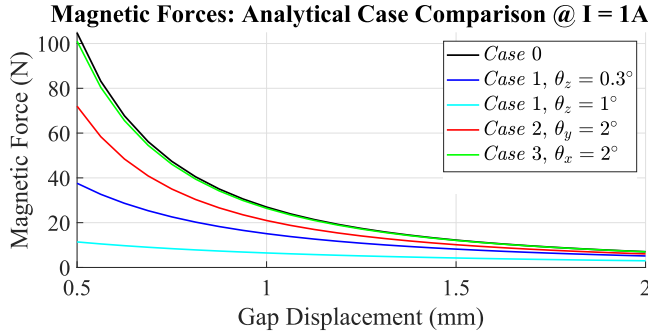


Fig. 11. Total analytical I-beam output force comparison of all considered cases at a current of $I = 1$ A and a 0.5–2 mm of gap separation [10].

can compound and show in the experimental results. As previously shown in Fig. 11, an increase in angular offset reduces the reluctance actuator's overall output force in all circumstances. Overall, the force could be approximated with reasonable precision by combining known FEA simulation results with derived analytical formulae. For upcoming studies, the force sensors and displacement gauge could be upgraded for better accuracy. To better reflect the real-world system, it would also be a consideration to use experimental data rather than FEA results for the 3D mapping correction factors. However, depending on the setup, experimental results may contain systematic errors.

The resulting force output shows the same trend throughout the current range which increases proportionally to the current squared ($F \propto i^2$) approximately. This goes with the analytical expression. Another takeaway from the results is how the magnitude of the force is

affected through the incorporation of a multi-axis offset in the mover. The results highlight that any rotational perturbations in the mover of the reluctance actuator can drastically affect the output performance and this effect should be characterized for high precision applications such as semiconductor lithography machines.

The output force of a reluctance actuator can be diminished based on unwanted axial offsets which are important to model for cases where the force output is of importance such as in high-precision motion control of lithography machines used in the semiconductor manufacturing industry. The results show how different cases of axial offsets can affect the force output and how multi-axis offsets can be characterized and predicted with an analytical equation. Another area where this could be used is when multi-axis offsets want to be taken advantage of for a rotational-based reluctance actuator that can be used in laser optic scanning applications. The analytical equation was able to match well with experimental findings. Future studies can use more sophisticated measurement techniques to reduce errors further. This paper shows the capabilities of utilizing force characterization of multi-axis offsets of the reluctance actuator. Understanding these effects can help in developing better control strategies in the future to combat or take advantage of the influence of multi-axis offsets of the C-core reluctance actuator.

7. Conclusion

To manufacture semiconductor devices more accurately and efficiently, the next development of the photolithography machine requires a more effective actuation system that can meet high acceleration and precision demands. Currently, Lorentz actuators are used due to

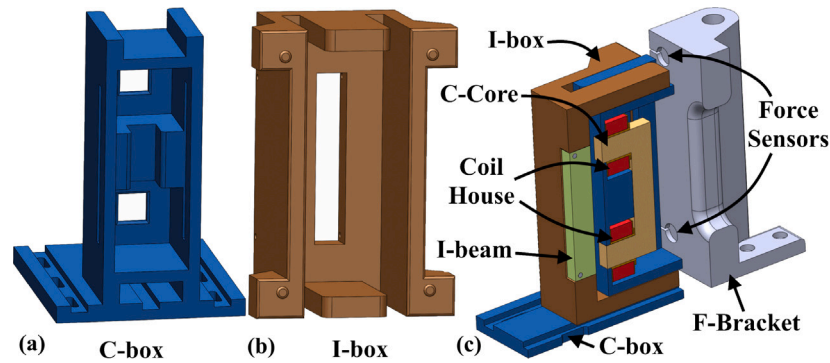


Fig. 12. The (a) C-box and (b) I-box house the C-core and I-beam respectively. The F-bracket (c) houses the force sensors and is what the I-box pushes against. Solidworks cross-sectional model (c) of the experimental assembly is also shown. All housing components are 3D printed [10].

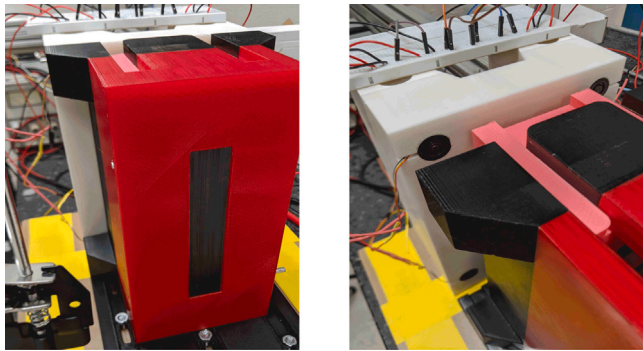


Fig. 13. The experimental setup is shown with the 3D printed components and force sensors that are connected to dSPACE.

their highly linear force-current dependence, making modeling and control straightforward. However, Lorentz actuators require effective cooling systems due to temperature and power restrictions. Reluctance actuators, on the other hand, can produce higher force and are quite effective, although their nonlinearity makes them more challenging to model. The force output can be dramatically altered by disturbances like asymmetries. This article highlights a successful technique for simulating the forces generated at multi-axial angular offsets in all degrees of rotational freedom. The main finding is how the force from different angular offset cases can be combined into a comprehensive model for the force output of the C-core reluctance actuator that considers multiple angular offsets in the mover element simultaneously. The experimental results demonstrate that errors can arise in the output force if there are angular perturbations in the mover of the reluctance actuator. Using the outlined method, these errors can be predicted and modeled for active control systems. Another application is when multi-axial offsets are beneficial for systems that require the mover to rotate in various degrees of freedom, such as laser scanning applications where the force and torque need to be modeled. The derived analytical model was found to match experimental findings within approximately 10%. These findings will improve the design of precision motion systems and their related control performance to eliminate asymmetrical air gap disturbances. Future research on control design will be used to enhance the motion control of reluctance-actuated systems.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

The authors acknowledge the financial support of the Natural Sciences and Engineering Research Council of Canada (NSERC), the European structural and investment funds, the Czech Ministry of Education, Youth and Sports through the Operational Programme Research, Development and Education, Project MS2014+: Mobility ČVUT - VTA, Project No. CZ.02.2.69/0.0/0.0/18_053/0016980.

Appendix. Polynomial correction equations

The polynomial correction functions are attached to the effective cross-sectional areas for the upper and lower air gaps of the reluctance actuator. The equations are found in (Appendix). The coefficient terms can be found in Table A.1. The equations are used in the total output force equation for the upper (11) and lower (12) air gaps of the reluctance actuator.

Table A.1

Coefficients for the polynomial correction functions on the upper α and lower air gaps β .

Term	Value	Term	Value	Units
α_1	0.9993	β_1	0.9891	–
α_2	0.07067	β_2	–0.1223	deg ^{–1}
α_3	–0.01683	β_3	0.02504	mm ^{–1}
α_4	–0.01796	β_4	0.1897	deg ^{–2}
α_5	–0.7009	β_5	–0.2571	deg ^{–1} mm ^{–1}
α_6	0.08285	β_6	0.02204	mm ^{–2}
α_7	–0.03938	β_7	–0.09227	deg ^{–3}
α_8	0.1231	β_8	0.02859	deg ^{–2} mm ^{–1}
α_9	0.339	β_9	0.1626	deg ^{–1} mm ^{–2}
α_{10}	–0.06827	β_{10}	–0.03298	mm ^{–3}
α_{11}	0.01575	β_{11}	0.03301	deg ^{–3} mm ^{–1}
α_{12}	–0.04931	β_{12}	–0.03544	deg ^{–2} mm ^{–2}
α_{13}	–0.05538	β_{13}	–0.02464	deg ^{–1} mm ^{–3}
α_{14}	0.01582	β_{14}	0.008659	deg ^{–4}
α_{15}	0.9837	β_{15}	0.9837	–
α_{16}	0.006107	β_{16}	0.006107	deg ^{–1}
α_{17}	0.07228	β_{17}	0.07228	mm ^{–1}
α_{18}	–0.00625	β_{18}	–0.00625	deg ^{–2}
α_{19}	0.008648	β_{19}	0.008648	deg ^{–1} mm ^{–1}
α_{20}	–0.03465	β_{20}	–0.03465	mm ^{–2}
α_{21}	0.000508	β_{21}	0.000508	deg ^{–2} mm ^{–1}
α_{22}	–0.00376	β_{22}	–0.00376	deg ^{–1} mm ^{–2}
α_{23}	0.005482	β_{23}	0.005482	mm ^{–3}
α_{24}	0.9812	β_{24}	0.9812	–
α_{25}	0.02393	β_{25}	0.02393	deg ^{–1}
α_{26}	0.07114	β_{26}	0.07114	mm ^{–1}
α_{27}	–0.01678	β_{27}	–0.01678	deg ^{–2}
α_{28}	0.01281	β_{28}	0.01281	deg ^{–1} mm ^{–1}
α_{29}	–0.02272	β_{29}	0.02272	mm ^{–2}

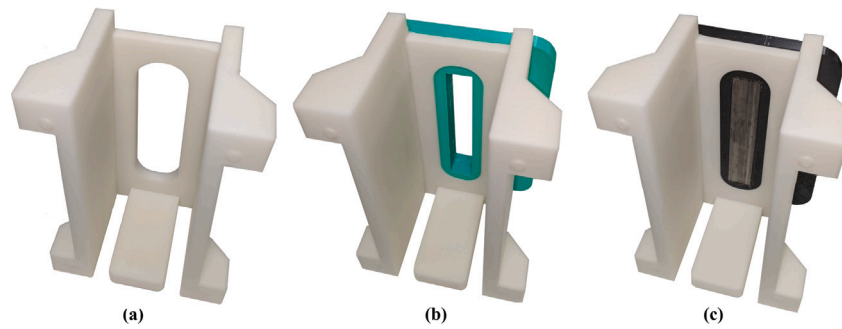


Fig. 14. The modular I-box design (a) allows for certain I-beam cases to be inserted into the back (b), making experimenting easier for each case. The I-beam can then be inserted (c) and held in place with pins.

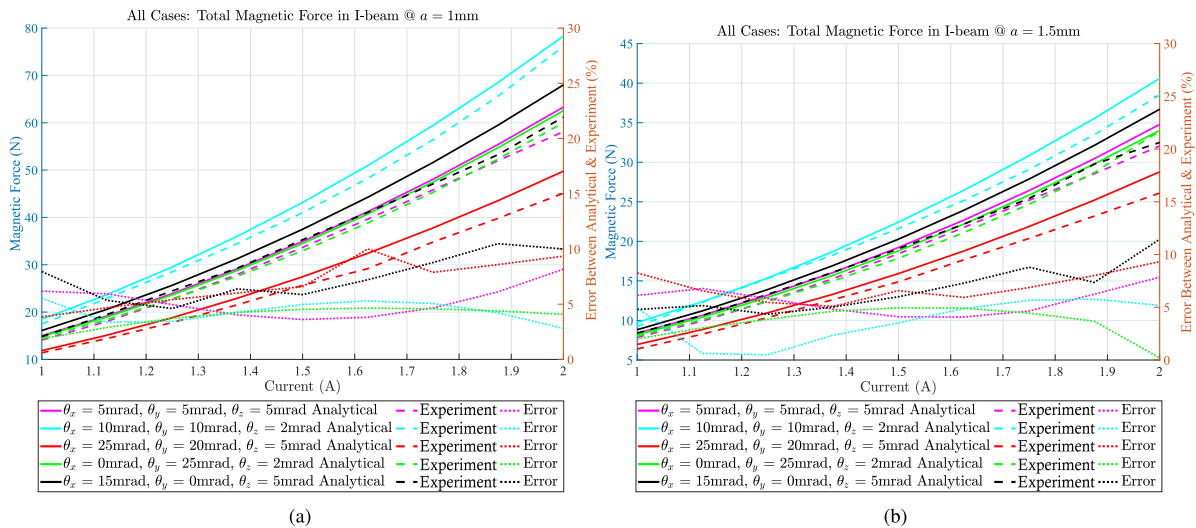


Fig. 15. The analytical and experimental results for the multi-axis Case 1 + 2 + 3, are shown for (a) $a = 1$ mm, (b) $a = 1.5$ mm to match with a maximum error of (a) 10.5%, (b) 11.4% for a range of 1–2 A of current.

$$A_1^*(\theta_x, \theta_y, \theta_z, a) = (\alpha_1 + \alpha_2\theta_z + \alpha_3a + \alpha_4\theta_z^2 + \alpha_5\theta_z a + \alpha_6a^2 + \alpha_7\theta_z^3 + \alpha_8\theta_z^2 a + \alpha_9\theta_z a^2 + \alpha_{10}a^3 + \alpha_{11}\theta_z^3 a + \alpha_{12}\theta_z^2 a^2 + \alpha_{13}\theta_z a^3 + \alpha_{14}a^4)(\alpha_{15} + \alpha_{16}\theta_y + \alpha_{17}a + \alpha_{18}\theta_y^2 + \alpha_{19}\theta_y a + \alpha_{20}a^2 + \alpha_{21}\theta_y^2 a + \alpha_{22}\theta_y a^2 + \alpha_{23}a^3)(\alpha_{24} + \alpha_{25}\theta_x + \alpha_{26}a + \alpha_{27}\theta_x^2 + \alpha_{28}\theta_x a + \alpha_{29}a^2)A_G$$

$$A_2^*(\theta_x, \theta_y, \theta_z, a) = (\beta_1 + \beta_2\theta_z + \beta_3a + \beta_4\theta_z^2 + \beta_5\theta_z a + \beta_6a^2 + \beta_7\theta_z^3 + \beta_8\theta_z^2 a + \beta_9\theta_z a^2 + \beta_{10}a^3 + \beta_{11}\theta_z^3 a + \beta_{12}\theta_z^2 a^2 + \beta_{13}\theta_z a^3 + \beta_{14}a^4)(\beta_{15} + \beta_{16}\theta_y + \beta_{17}a + \beta_{18}\theta_y^2 + \beta_{19}\theta_y a + \beta_{20}a^2 + \beta_{21}\theta_y^2 a + \beta_{22}\theta_y a^2 + \beta_{23}a^3)(\beta_{24} + \beta_{25}\theta_x + \beta_{26}a + \beta_{27}\theta_x^2 + \beta_{28}\theta_x a + \beta_{29}a^2)A_G$$

References

- [1] Butler H. Position control in lithographic equipment. *IEEE Control Syst Mag* 2011;31:28–47.
- [2] Valentin CL. Curvature manipulation of photomasks - enhancing the imaging performance of immersion lithography equipment (Ph.D. thesis), Dept. Precision and Microsystems Engineering. TU Delft, Delft University of Technology, Delft, Netherlands; 2013.
- [3] Otsu Y, Somaya K, Yoshimoto S. High-speed stability of a rigid rotor supported by aerostatic journal bearings with compound restrictors. *Tribol Int* 2011;44(1):9–17.
- [4] Dupont R. Robust rotor dynamics for high-speed air bearing spindles. *Precis Eng* 2015;40:7–13.
- [5] Ye Y, Chen X, Hu Y, Luo X. Effects of recess shapes on pneumatic hammering in aerostatic bearings. *Proc Inst Mech Eng J* 2010;224(3):231–7.
- [6] Aoyama T, Koizumi K, Kakinuma Y, Kobayashi Y. Numerical and experimental analysis of transient state micro-bounce of aerostatic guideways caused by small pores. *CIRP Ann* 2009;58(1):367–70.
- [7] MacKenzie I. Design and control methods for high-accuracy variable reluctance actuators (Ph.D. thesis), Massachusetts Institute of Technology; 2015.
- [8] Okay A, Erkorkmaz K, Khamesee MB. Mechatronic design, actuator optimization, and control of a long stroke linear nano-positioner. *Precis Eng* 2018;52:308–22.
- [9] Zhang X, Lai L, Zhang L, Zhu L. Hysteresis and magnetic flux leakage of long stroke micro/nanopositioning electromagnetic actuator based on Maxwell normal stress. *Precis Eng* 2022;75:1–11.
- [10] Pumphrey M, Alatawneh N, Janaideh M. Modeling and analysis of reluctance motion system with asymmetrical air gaps. *Rev Sci Instrum* 2022;93:075001.
- [11] Vrijsen N, Jansen J, Lomonova E. Comparison of linear voice coil and reluctance actuators for high-precision applications. 2010, p. S3–29.
- [12] van Zundert J, Oomen T, Verhaegh J, Aangenent W, Antunes DJ, Heemels WPMH. Beyond performance/cost tradeoffs in motion control: A multirate feedforward design with application to a dual-stage wafer system. *IEEE Trans Control Syst Technol* 2018;28(2):1–14.
- [13] Burgstaller I, Ito S, Fujimoto H, Schitter G. Development of reluctance actuator for high-precision positioning and scanning motion. In: 2021 IEEE international conference on mechatronics (ICM). 2021, p. 1–6.
- [14] Navarrete MO, Heertjes MF, Schmidt RHM. Common zeros in synchronization of high-precision stage systems. In: IEEE international conference on mechatronics (ICM), Nagoya, Japan. 2015, p. 602–7.
- [15] Oomen T, van Herpen R, Quist S, van de Wal M, Bosgra O, Steinbuch M. Connecting system identification and robust control for next-generation motion control of a wafer stage. *IEEE Trans Control Syst Technol* 2013;22(1):102–18.
- [16] Heertjes M, Hennekens D, Steinbuch M. MIMO feed-forward design in wafer scanners using a gradient approximation-based algorithm. *Control Eng Pract* 2010;18(5):495–506.
- [17] Baggen M, Heertjes M, Kamidi R. Data-based feed-forward control in MIMO motion systems. In: American control conference, Seattle, WA, USA. 2008, p. 3011–6.
- [18] Cigarini F, Csencsics E, Schlarp J, Ito S, Schitter G. Multiphysics finite element model for the computation of the electro-mechanical dynamics of a hybrid reluctance actuator. *Math Comput Model Dyn Syst* 2020;26(4):322–43.

- [19] Ito S, Cigarini F, Schitter G. Flux-controlled hybrid reluctance actuator for high-precision scanning motion. *IEEE Trans Ind Electron* 2020;67(11):9593–600.
- [20] Katalenic A, Butler H, van den Bosch PPJ. High-precision force control of short-stroke reluctance actuators with an air gap observer. *IEEE/ASME Trans Mechatronics* 2016;21(5):2431–9.
- [21] Long Y, Huang S, Peng L, Wang S, Zhao W. A novel compensation method of probe gesture for magnetic flux leakage testing. *IEEE Sens J* 2021;21(9):10854–63.
- [22] Moya-Lasheras E, Sagues C. Run-to-run control with Bayesian optimization for soft landing of short-stroke reluctance actuators. *IEEE/ASME Trans Mechatronics* 2020;PP:1.
- [23] Liu Y-P, Liu K-Z, Xiaofeng Y. Nonlinear current control for reluctance actuator with hysteresis compensation. *J Control Sci Eng* 2014;2014.
- [24] Xu Y, Shu F, Yang X, Su X, Hu B. A new regressive RBF neural network model for rate-dependent hysteresis in reluctance actuators. In: 2021 13th international symposium on linear drives for industry applications (LDIA). IEEE; 2021, p. 1–6.
- [25] Katalenic A, Boeij J, Butler H, Bosch P. Linearization of a current-driven reluctance actuator with hysteresis compensation. *Mechatronics* 2013;23:163–71.
- [26] Vrijnsen N, Jansen J, Compter J, Lomonova E. Measurement method for determining the magnetic hysteresis effects of reluctance actuators by evaluation of the force and flux variation. *Rev Sci Instrum* 2013;84:075003.
- [27] Moya-Lasheras E, Sagues C, Llorente S. An efficient dynamical model of reluctance actuators with flux fringing and magnetic hysteresis. *Mechatronics* 2021;74:102500.
- [28] Ramírez-Laboreo E, Roes M, Sagues C. Hybrid dynamical model for reluctance actuators including saturation, hysteresis and eddy currents. *IEEE/ASME Trans Mechatronics* 2019;24:1396–406.
- [29] Pechgraber D, Csencsics E, Schitter G. Resonant rotational reluctance actuator for large range scanning mirrors. *IEEE/ASME Trans Mechatronics* 2023;1–10.
- [30] Al Saaideh M, Alatawneh N, Al Janaideh M. Multi-objective optimization of a reluctance actuator for precision motion applications. *Journal of Magnetism and Magnetic Materials* 2022;546:168652.
- [31] Al Saaideh M, Boker AM, Al Janaideh M. Output-feedback control of electro-magnetic actuated micropositioning system with uncertain nonlinearities and unknown gap variation. In: 2022 IEEE 61st Conference on Decision and Control (CDC). IEEE; 2022, p. 2481–6.